# Rebecca Kyer

rkyer@msu.edu • 0009-0006-2411-5162 • rkyer

## Education

**Michigan State University** **East Lansing, MI**
Ph.D. Candidate; thesis advisor: Jay Strader *Nov 2024 – present*

M.S., Astrophysics and Astronomy *Aug 2022 – Nov 2024*

**University of Washington** **Seattle, WA**
B.S., Physics and Astronomy *Jun 2021*
*Astronomy Departmental Honors*

## Past Research Appointments

**Green Bank Observatory** **Green Bank, WV**
Graduate Student in Residence *May 2026*

## Publications

**Refereed**

5. "Investigating Four New Candidate Redback Pulsars Discovered in the Image Plane." Petrou, F., Hurley-Walker, N., McSweeney, S. et al. (incl. **Kyer, R.**) 2025, PASA, 42, 139

4. "Multiwavelength Evidence for Two New Candidate Transitional Millisecond Pulsars in the Subluminous Disk State: 4FGL J0639.1–8009 and 4FGL J1824.2+1231." **Kyer, R.**, Roy, S., Strader, J., et al. 2025, ApJ, 983, 112

3. "A Survey for Radio Emission from White Dwarfs in the VLA Sky Survey." Pelisoli, I., Chomiuk, L., Strader, J., et al. (incl. **Kyer, R.**) 2024, MNRAS, 531, 1805

2. "Monitoring the X-ray Variability of Bright X-ray Sources in M33." **Kyer, R.**, Albrecht, S., Williams, B. F., et al. 2024, ApJ, 961, 168

1. "Revisiting the Classics: On the Evolutionary Origin of the 'Fe II' and 'He/N' Spectral Classes of Novae." Aydi, E., Chomiuk, L., Strader, J. et al. (incl. **Kyer, R.**) 2024, MNRAS, 527, 9303

**Non-refereed**

"The Advanced X-ray Imaging Satellite Community Science Book." Koss, M., Aftab, N., Allen, S. et al (incl. **Kyer, R.**) 2025, arXiv:2511.00253

"Unlocking the Rapid X-ray Variability of Transitional Millisecond Pulsars." **Kyer, R.**, Ng, M., Coti Zelati, F. et al.

"Catching Spiders with AXIS." Coti Zelati, F., An, H., Baglio, M. C. et al. (incl. **Kyer, R.**)

Transient Name Server Classification Report No. 2024-2571, classification of SN 2024pxg as a SN II. Bostroem, K. A., **Kyer, R.**, Strader, J. et al. 2024

Various classification reports in The Astronomer's Telegram (No. 16583, 16440, 16294)

## Telescope Time

*XMM-Newton* AO-25 ID 098076, 35 ksec awarded, Scientific PI *2025*

*Green Bank Telescope* 2026A GBT26A-226, 15 hr awarded/4.5 hr observed, PI/Observer *2025*

*Gemini-North* Fast Turnaround ID GN-2025B-FT-202, 0.38 hr, PI *Aug 2025*

*Green Bank Telescope* 2025B Legacy ID QS402, 15 hr, Co-I/Observer *2025*

*XMM-Newton* AO-24 ID 096163, 57 ksec awarded, Scientific PI *2025*

*Swift* Target of Opportunity ID 21032, 8.5 ksec, PI *Aug 2024*

*Swift* Target of Opportunity ID 20572, 3.8 ksec, PI *June 2024*

190+ hours remote observing with the 4.1-m *SOAR* telescope *2023–present*

## Research Advising Experience for MSU Undergraduate Students

Jadyn Elster *May 2026–present*

Learning to analyze optical spectra from long-term monitoring of an evolved massive star with a hidden compact object companion. Measurements made by Jadyn will the basis for a publication on the results of this monitoring data.

Andrew Harms, co-advised with Ryan Urquhart *May 2025–present*

Learned to combine archival X-ray and radio data to constrain the prevalence of hundreds-of-pc-scale radio/optical nebulae associated with ultraluminous X-ray sources in the Southern Hemisphere sky to predict yield from future deep radio surveys. Made an independent discovery of a new nebula and learned to analyze a follow-up optical spectrum. Preparing manuscript for first-author publication of both results.

Subhroja Roy (now U. Notre Dame Nuclear Astrophysics Ph.D. student) *May 2024–May 2025*

Learned to combine archival $\gamma$-ray, X-ray and optical data sets to identify potential spider MSPs and analyze follow-up optical spectroscopy. Completed astronomy senior thesis presenting two new strong candidate binary MSPs, one of which has been published with Subhroja as second author and the other is to be included in an upcoming publication.

## Research Presentations

Green Bank Observatory (GBO) Science Lunch talk (Green Bank, WV) *May 2026*

GBO/NRAO Radio Astronomy Bootcamp science talk (Green Bank, WV) *May 2026*

MSU Astronomy Seminar talk (East Lansing, MI) *Jan 2026*

AAS 247 AXIS Special Session invited talk (Phoenix, AZ) *Jan 2026*

AAS 247 Accretion and Evolution in X-ray Binary Systems talk (Phoenix, AZ) *Jan 2026*

Compact Objects in Michigan and Ontario (COMO) talk (Flint, MI) *Sept 2025*

Vasto Accretion Meeting talk (Vasto, Italy) *June 2025*

AXIS Community Science Meeting talk (Annapolis, MD) *May 2025*

Int'l Symposium of Science with the SOAR Telescope talk (East Lansing, MI) *May 2025*

ESAC XMM-Newton Science Workshop poster (Madrid, Spain) *June 2024*

COMO talk (Dearborn, MI) *May 2024*

## Outreach Activities

| | |
|---|---|
| Letters to a Pre-Scientist Penpal | *2 penpals* |
| Astronomy on Tap Speaker (Lansing, MI) | *Nov 2023, Apr 2025* |
| Campus Observatory Public Night Volunteer (East Lansing, MI) | *Aug 2022–present* |

## Professional Service

| | |
|---|---|
| ESA NewAthena Science Community Working Group Member | *Feb 2026–present* |
| AAS 247 Chambliss Poster Competition Judge | *Jan 2026* |
| NASA AXIS Probe CO/SNR Working Group Member | *Dec 2024–Mar 2026* |
| P-A Research Experiences for Drew Scholars Junior Director | *Apr 2024–present* |
| Faculty Search Committee Graduate Student Representative | *Feb 2024* |
| Stellar Mentorship Program Coordinator and Peer Mentor | *Aug 2023–present* |
| Astro Coffee Journal Club Organizer | *Jun 2023–present* |

## Teaching Experience

| | |
|---|---|
| Teaching Assistant AST 208: Planets and Telescopes (30 students; MSU) | *Jan–May 2024* |
| Teaching Assistant ISP 205L: Visions of the Universe (400 students; MSU) | *Aug 2022–Dec 2023* |
| Instructor GEN ST 199: Astronomy First-year Interest Group (30 students; UW) | *Sep–Dec 2020* |

## Awards and Honors

| | |
|---|---|
| Honorable Mention, NSF Graduate Research Fellowship | *Apr 2024* |
| Graduate Teaching Assistant Award, MSU Physics and Astronomy | *Apr 2024, Apr 2023* |
| Mervyn Mork Excellence in Teaching Award, MSU College of Natural Science | *Nov 2023* |
| Baer Prize for Excellence in Research, Academics and Outreach, UW Astronomy | *May 2021* |